

Ragib Sakib

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EDUCATION

University of Southern California | Los Angeles, CA | *August 2026 - June 2028 (Expected)*

Master of Science (MS) in **Computer Science (Artificial Intelligence)**

University of California, Irvine | Irvine, CA | *September 2020 - June 2024*

Bachelor of Science (BS) in **Software Engineering**, GPA: **3.6**

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, Rust, Java

Machine Learning & AI: PyTorch, TensorFlow, OpenCV, deep learning, computer vision, recommendation systems

ML Systems & Data: FAISS, sentence-transformers, ChromaDB, SQL, SQLite, model inference, RAG

Applications & Platforms: FastAPI, Docker, Google Cloud Run, Gradio, React, Ionic, React Native, Linux, Git

WORK EXPERIENCE

Canon USA

Software and Machine Learning Engineer | Irvine, CA | *September 2023 - February 2025*

- Designed a Python vision pipeline using SAM 2, Hough transforms, and clustering for object corner localization
- Ported spatial clustering and geometric algorithms to C++, reducing execution latency by 90% via pass-by-reference
- Integrated ML runtimes into production software, enabling responsive user experiences through low-latency inference
- Shipped a cross-platform Ionic React application to iOS and Android from one shared codebase to expand user reach

Canon USA

Software Intern | Irvine, CA | *June 2023 - August 2023*

- Modernized an object-oriented C# codebase from legacy UWP, reducing technical debt and improving maintainability
- Built a cross-platform mobile and web application that expanded the project team's existing web-only client
- Collaborated in a fast-paced Agile workflow using Jira for planning, implementation, testing, and delivery

Boeing Satellite Systems

Engineering Intern | El Segundo, CA | *June 2019 - August 2019*

- Led a quality engineering team in an intern competition before 300 engineers, placing top three in every category
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SOFTWARE PROJECTS

LoL Team Recommender | *September 2026*

[DEMO](#) | [GITHUB](#)

- Trained a PyTorch two-tower neural model on 69,378 player pairs to rank teammates by role, rank, and preferences
- Designed a joint neural lineup scorer using player interactions and champion pools to optimize four selections together
- Built an NLP and RAG pipeline with Sentence Transformers, ChromaDB, and Qwen to explain 2,854 player profiles
- Served model inference through a Dockerized Gradio/FastAPI application on Google Cloud Run with scale-to-zero
- Benchmarked role-and-rank retrieval against a popularity baseline, improving recommendation NDCG@10 by 29%

MovieLens 1M Two-Tower Recommendation Engine | *June 2026*

[DEMO](#) | [GITHUB](#)

- Built a PyTorch two-tower neural retrieval model with in-batch negative ranking loss for movie recommendations
- Engineered a user tower from demographics and sequential watch history to capture personalized preference signals
- Precomputed and indexed movie embeddings in FAISS for efficient, low-latency model inference across the catalog
- Visualized learned embeddings with t-SNE to validate genre-based clustering in the representation space

MUGIC Gait Analysis Software (College Capstone) | *September 2023 - June 2024*

Project Lead/Front-End Engineer and Designer

- Led 3 programmers and 2 designers to build a cross-platform rehabilitation application for an Irvine-based start-up
- Integrated rehabilitation motion-sensor data to deliver meaningful feedback on each user's walking gait
- Improved software quality with Python linting, 80% unit-test coverage, and comprehensive design documentation